

Im2Frac - Development of an image-to-analysis pipeline for the modeling of fracture processes

Special Project SoSe 2026

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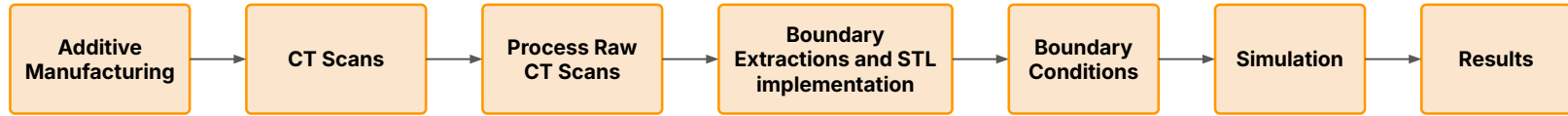
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Software!

- Existing Software to deal with CT Scans: Slicer 3D
- Problem:** Developed for Medical CT scans

Python Scripts to GUI

- Core Tech Stack: PyQt6 (GUI), VTK (3D View)
- Dependencies: Nibabel, Numpy, Pandas, Matplotlib, psutil and more!

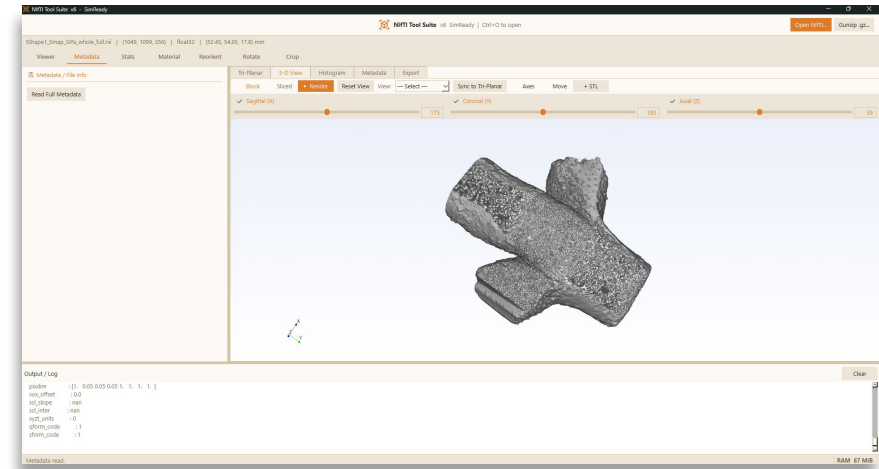


Figure 1.1: Custom CT-analysis GUI (PyQt6/VTK) rendering a 3D view of an additively manufactured T-shape specimen

Processing Raw CT Scans

- DICOM (.dcm) to NifTi (.nii)
- Crop to reduce the size

Removing Background Air

- Otsu's method [1]

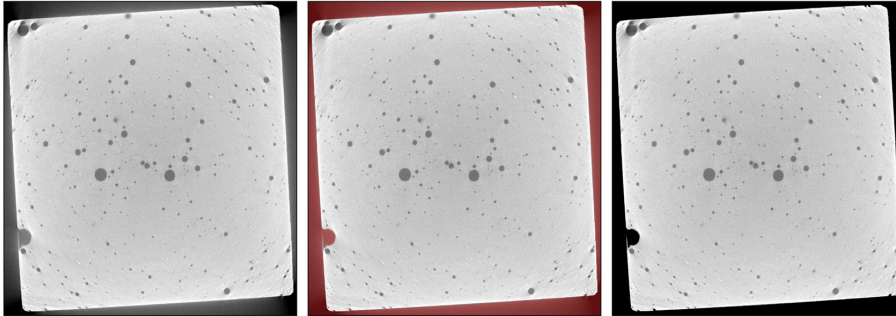


Figure 1.2 : Background/air removal on a cube cross-section: raw slice, detected background (red), cleaned slice

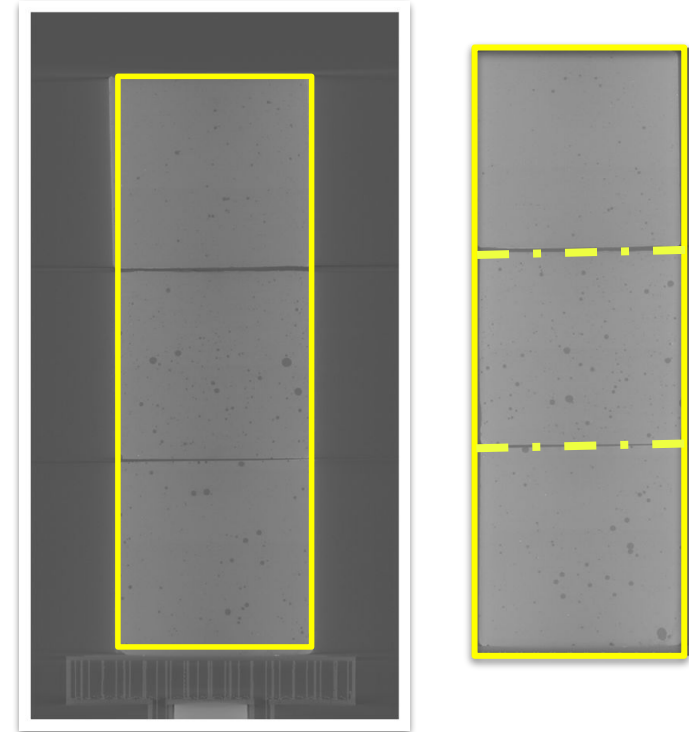


Figure 1.3 : Crop region and valid specimen-height range (dashed boundaries) to extract cubes

Processing Raw CT Scans: Beam Hardening Artifact

$$\text{cupping index} = \frac{(\text{brightness near edge} - \text{brightness near centre})}{\text{brightness near edge}} \times 100\%$$

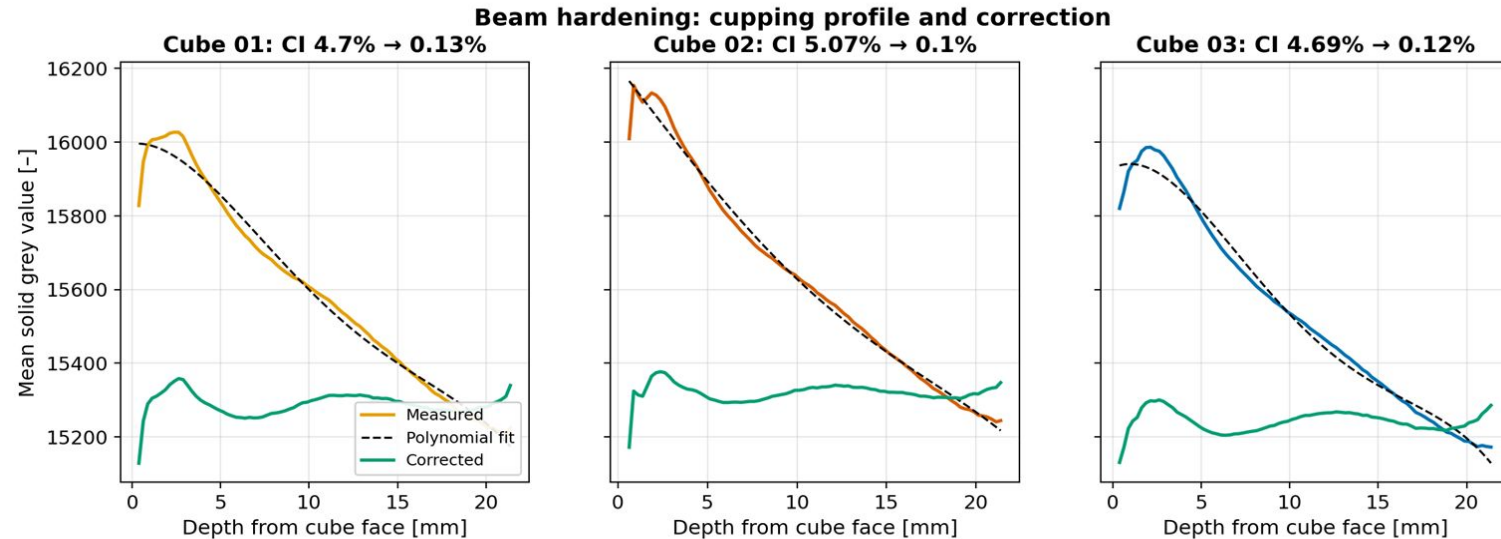
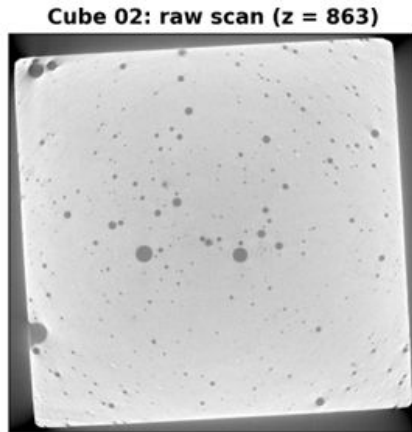


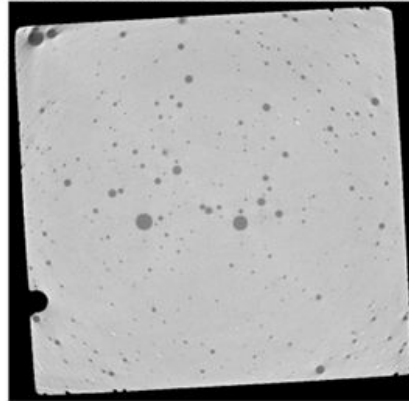
Figure 1.4 : Cupping profile and polynomial correction for Cubes 01-03 (measured, fit, corrected) [2][3][4]

Processing Raw CT Scans: Beam Hardening Artifact

$$\text{corrected brightness} = \frac{\text{original brightness} \times \text{reference brightness}}{\text{fitted curve value at that depth}}$$



Corrected volume (same slice, same grey scale)



Brightness profile — cupping index 5.07% → 0.1%

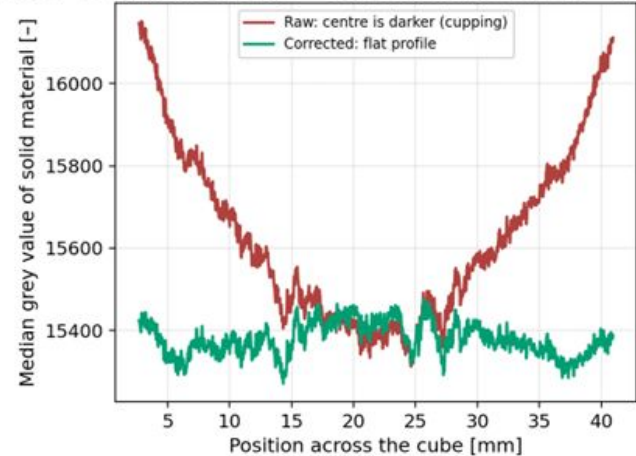


Figure 1.5 : Cube 02: raw vs. corrected slice and brightness-profile comparison (cupping index 5.07% → 0.1%)

Pores Segmentation

- Pores are where cracks start and its accuracy directly drives simulation validity

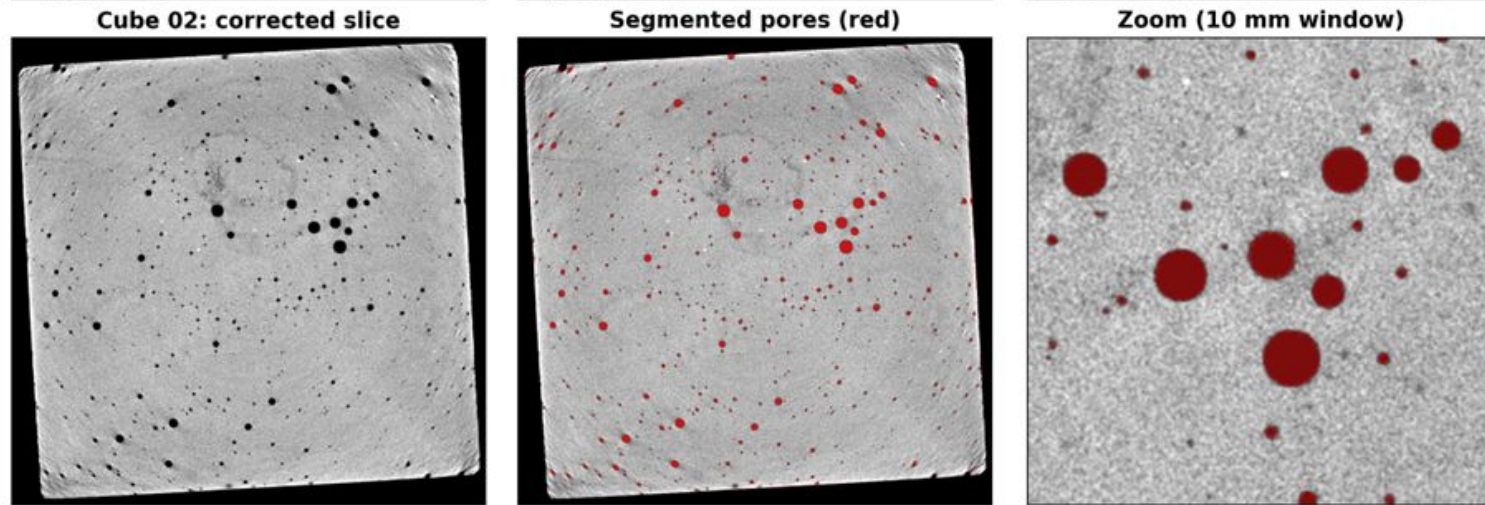
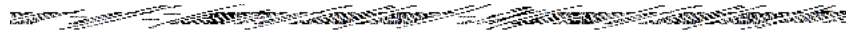


Figure 1.6 : Cube 02: corrected slice, segmented pores (red overlay), and 10 mm zoom on the densest pore region [5][6]

Quantifying Pores and Deducing RVE's

pores (red) T=11157 | porosity=3.31%

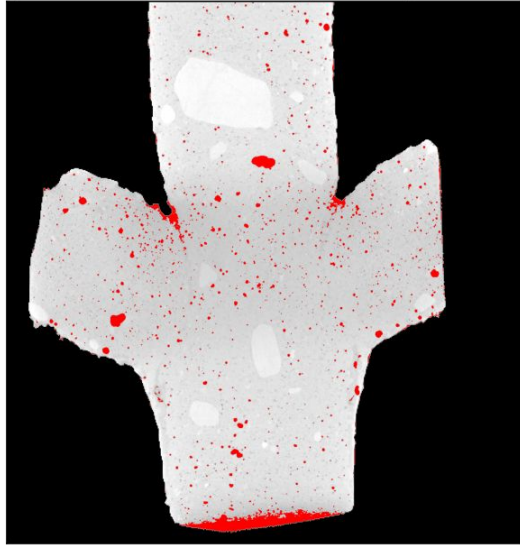


Figure 1.7 : Another example of Pore Segmentation

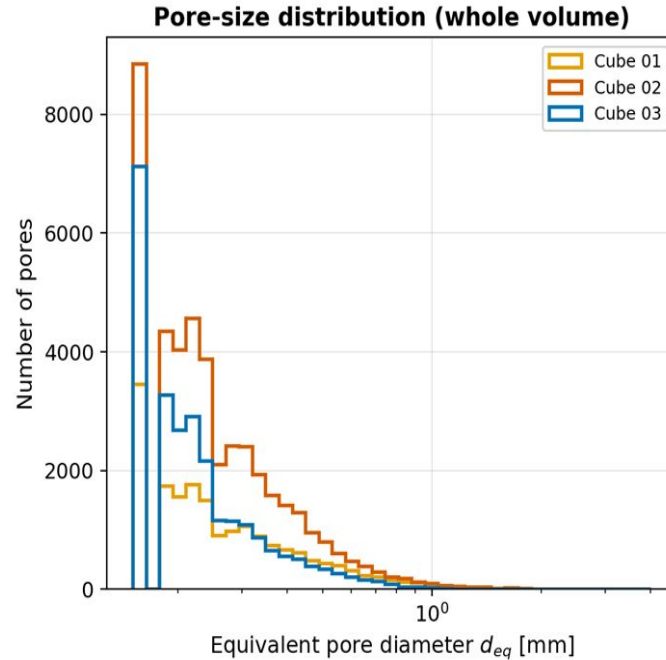
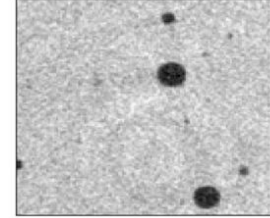
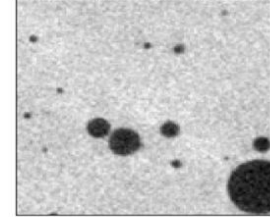


Figure 1.8 : Pore-size distribution (equivalent diameter) for Cubes 01-03 [7]

Cube 01: grey values (6.4 mm)



Cube 02: grey values (6.4 mm)



Cube 03: grey values (6.4 mm)

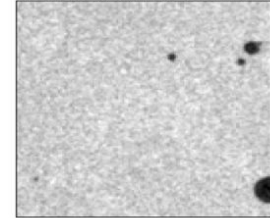


Figure 1.9 : 6.4 mm sub-volume grey-value samples per cube, used for the RVE

Young's Modulus Mapping:

Relative Mapping:

- Solid = 1
- Void = 0

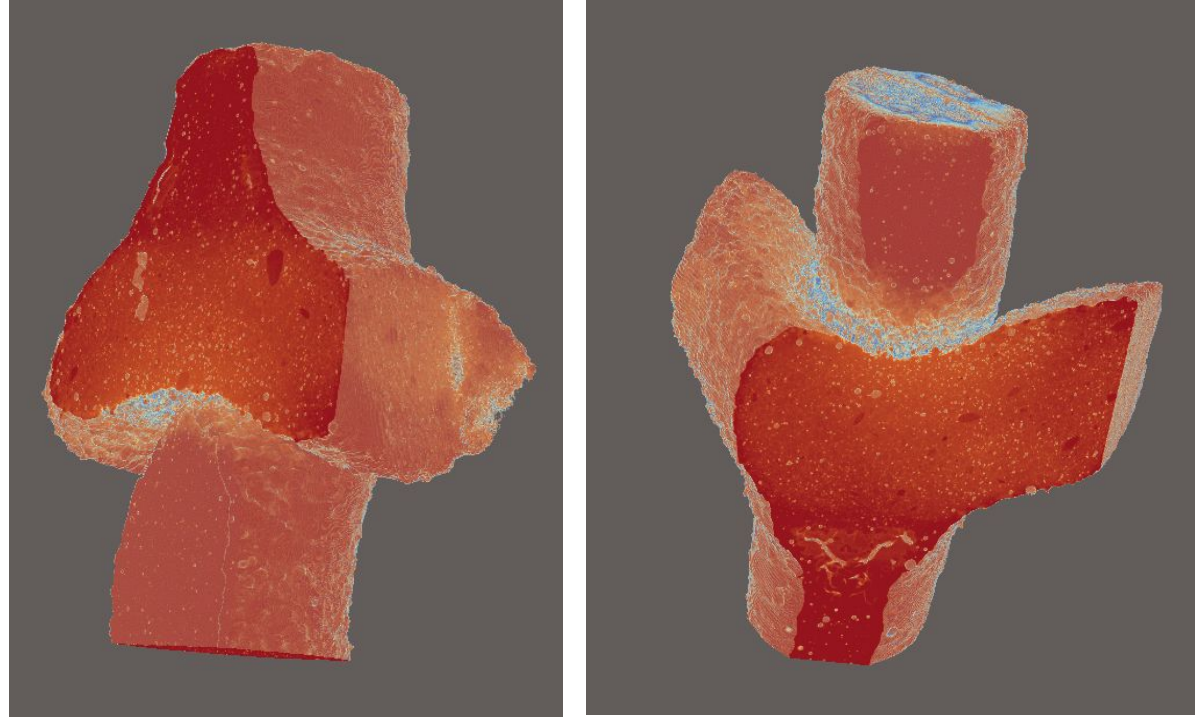
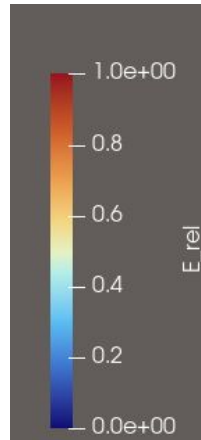


Figure 1.10 : Relative Young's-modulus field (E_{rel} , solid = 1 / void = 0) on the T-Shape specimens

Greyscale → Young's Modulus Mapping

- Goal: To match every voxel to a Young's Modulus value, such that simulation is natural
- Why was it dropped?
 - Complex research topic and difficult to be defended scientifically
 - No phantom calibration for the CT scans to proceed
 - Different phases compared to bones
- Next:
 - Pass the processed, corrected and refactored CT scans and its RVE's for boundary extractions and STL creation for simulation

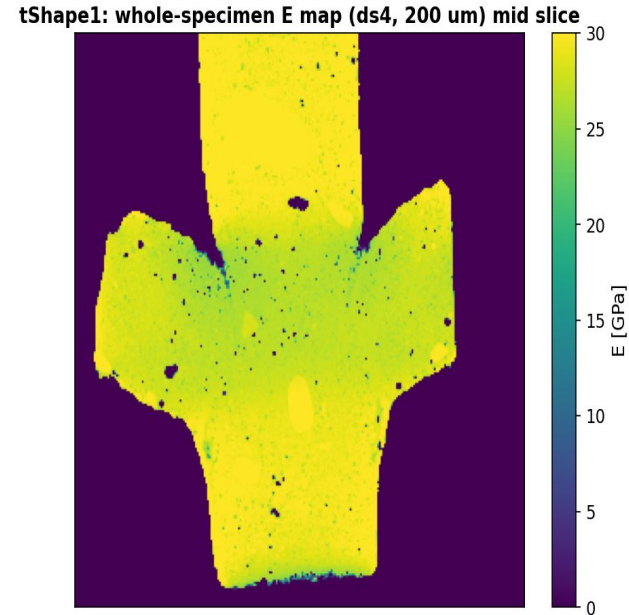
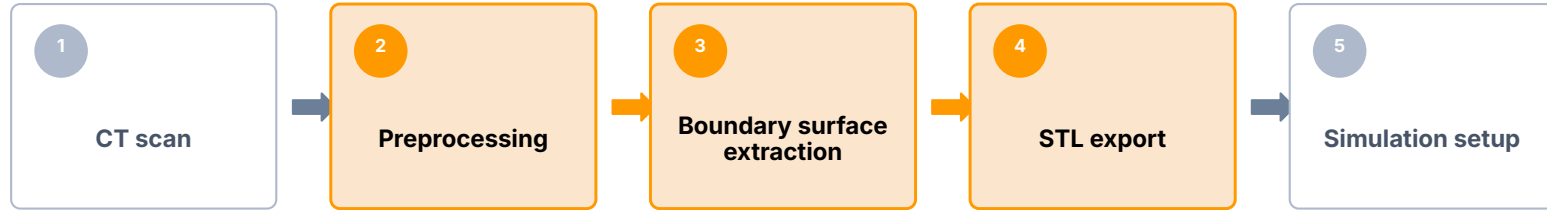


Figure 1.11 : Example of Young's Modulus Mapped CT Scans [10]

From CT Data to Simulation-Ready Inputs



Input

NIfTI volume from CT data

Voxel data + physical coordinate information

Task

Prepare image + identify regions

Cropping, orientation, ROI, threshold, surface extraction

Output

STL markers for boundary conditions

Used to apply load/support surfaces in simulation

Why Preprocessing Is Necessary

- **Isolate the specimen**

Cropping removes unnecessary regions and keeps the volume focused on the specimen.

- **Standardize geometry**

Orientation, voxel spacing and physical coordinates are checked so that the geometry remains consistent.

- **Prepare material information**

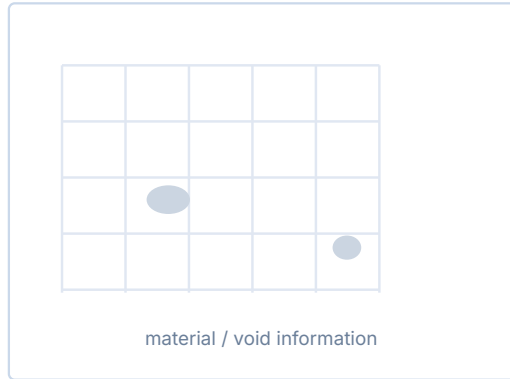
Thresholding supports material/void masks used later by the surface-extraction step.

- **Keep simulation alignment**

The prepared NIfTI, exported STL surfaces and solver domain must share the same scale.

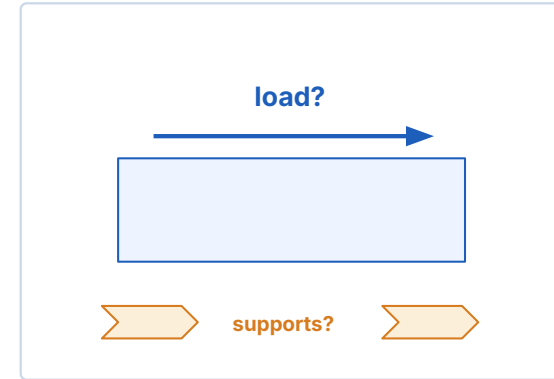
Problem: The CT Volume Has Geometry, Not Instructions

What the NIfTI already gives us



needs labels

What is still missing



- NIfTI represents the specimen geometry and voxel information
- Boundary conditions must be assigned to selected surfaces
- Manual STL creation can cause coordinate mismatch or wrong placement

Solution: Extract Boundary-Condition Surfaces from the NIfTI

- Select a local region on the CT volume, find the exposed material surface in one direction, and export only that surface as an STL marker.



Voxel corners are converted to physical millimetre coordinates before writing the STL. This keeps the exported surface in the same coordinate system as the NIfTI volume and the simulation domain.

GUI Workflow: Select, Export, Inspect

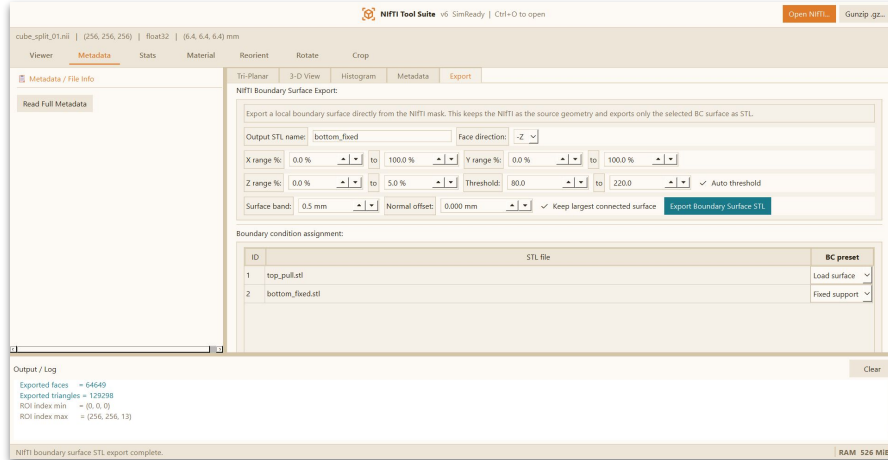


Figure 2.1 : Boundary-surface export controls for defining face direction, ROI, threshold and cleanup settings.

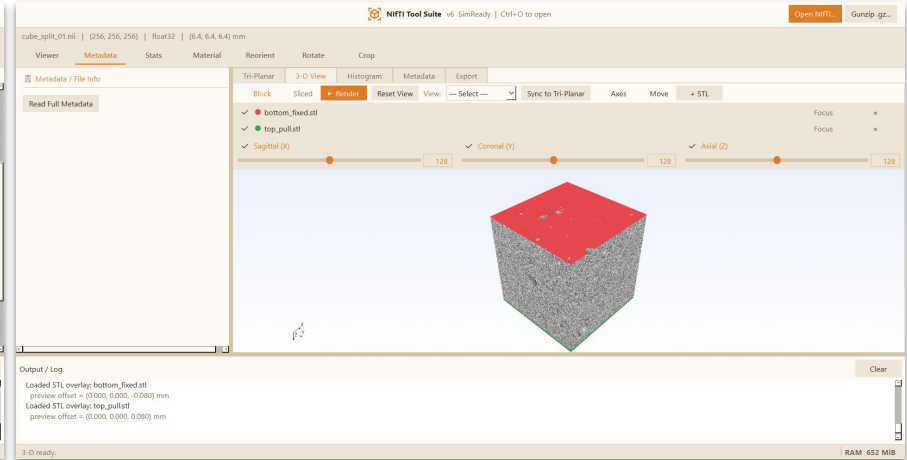


Figure 2.2 : 3D preview of the exported STL surface overlaid on the CT-derived specimen.

1

Choose face direction

+X / -X / +Y / -Y / +Z / -Z

2

Set local region

percentage ranges for X, Y, Z

3

Create clean surface

threshold + surface band + largest component

4

Verify visually

overlay STL on the CT volume

Validation: Make Sure the STL Is Usable

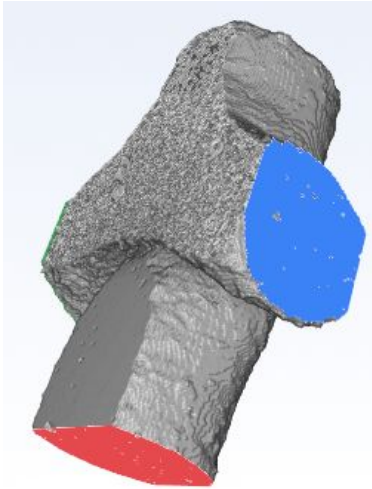


Figure 2.3 : Visual validation of the exported boundary-condition surfaces on the T-shaped specimen.

Validation checks

Visual overlay

Does the STL lie on the selected specimen surface?

Coordinate bounds

Are STL coordinates inside the NIfTI physical domain?

Direction check

Is the exported face on the intended side?

Solver intersection

Does the simulation actually apply the load/support?

The STL can look correct in the viewer, but the final confirmation is whether the C++ simulation detects the boundary region and applies the condition correctly.

Output: A Cleaner Handoff to Simulation

Outputs from this stage

1. Preprocessed NIfTI

specimen geometry and voxel information

2. Boundary STL files

selected surfaces for loads and supports

3. Export metadata / JSON

paths, roles, thresholds and physical bounds



Next stage uses them to

- Embed the voxel geometry in the FCM background mesh
- Apply loads and supports using the STL markers
- Solve the fracture problem and compare with experiment

Boundary Conditions in FCM - The Core Challenge

What is a Boundary Condition?

Defines what happens at specimen surface

- What happens at the specimen surface during loading
- Dirichlet - prescribes displacement
- Neumann - prescribes force
- *For fracture, displacement-controlled loading*

Why is it hard in FCM?

Standard FEM	Finite Cell Method
Boundary = mesh boundary	Boundary inside the grid
Apply BC directly	Special treatment needed

Real CT-scan geometries need a new approach - and that is where the STL's come in.

STL Surfaces as Boundary Conditions

WHERE - STL file

- Triangulated surface mesh defining a surface in 3D space
- Placed by the user on the specimen via the GUI
- Defines exact location + shape of the BC surface
- e.g. flat patch at the base (support), patch on an arm (load)

WHAT - JSON config

- Stores BC metadata per surface
- Type: Dirichlet (displacement) or Neumann (force)
- Constrained axes: $X=0$, $Y=1$, $Z=2$
- Values: $[0,0,0]$ fixed; $[0, -1.0, 0]$ downward push

HOW - C++ penalty method

- Reads STL surface + JSON metadata
- Integrates over STL triangles inside the FCM grid
- Adds penalty stiffness terms to the global matrix
- Pulls displacement toward target at each integration point

Generic JSON-Driven BC System

Why JSON?

- Separates configuration from code
- Simulation code is completely generic; no per-test hardcoding
- GUI writes the JSON and the cpp script reads it

JSON structure

```
{
  "specimen": {
    "nifti_file": "tShape1_Emap_GPa.nii",
    "E_ref": 30.0, "nu": 0.2,
    "void_threshold": 0.4746
  },
  "boundary_conditions": [
    { "name": "bottom_fixed",
      "stl_file": "case2_bottom_fixed.stl",
      "simulation_definition": {
        "constrained_axes": [0, 1, 2],
        "displacement_values": [0, 0, 0],
        "is_load_surface": false } },
    { "name": "sideA_push",
      "stl_file": "case2_sideA_push.stl",
      "simulation_definition": {
        "constrained_axes": [1],
        "displacement_values": [-1.0],
        "is_load_surface": true,
        "measure_reaction_force": true } }
  ]
}
```

Adding a new BC surface = one new JSON entry. The C++ code requires zero changes.

Validation: Cantilever Bending and 3 - Point Bending

- Dogbone specimen with two STL surfaces - one fixed end, one displaced end
- Confirmed: displacement correctly enforced via STL-based penalty method

Specimen	E (kN/mm ²)	Result	Residual
Dogbone cantilever	30	Correct deformation	9.01e-07

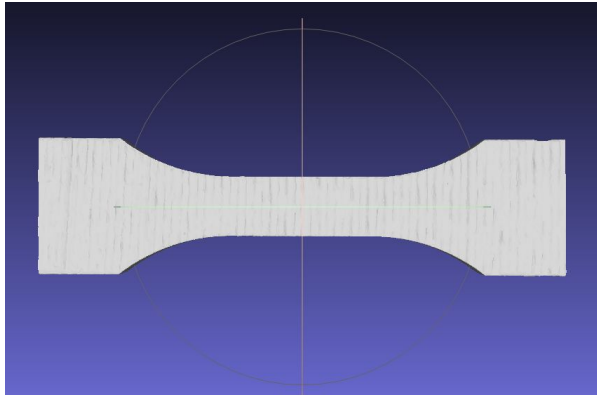


Figure 3.1: The Dogbone geometry

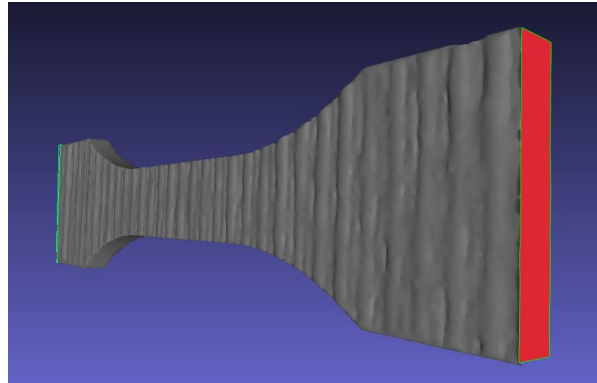


Figure 3.2: Right fixed support (in red)

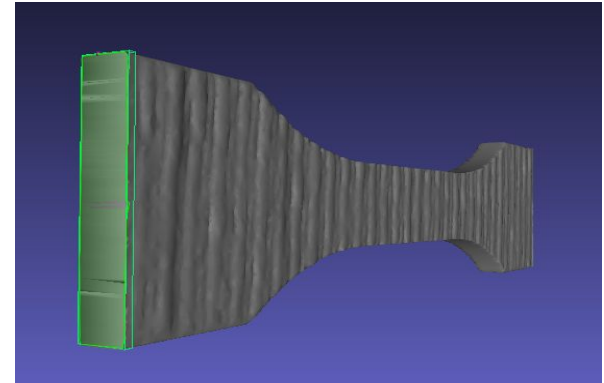


Figure 3.3: To be Displaced End (in green)

Validation: Dogbone Cantilever Bending

- RHS fixed · LHS displaced -1 mm with a residual, $9.01\text{e}-07$
- Displacement correctly enforced via STL-based penalty method

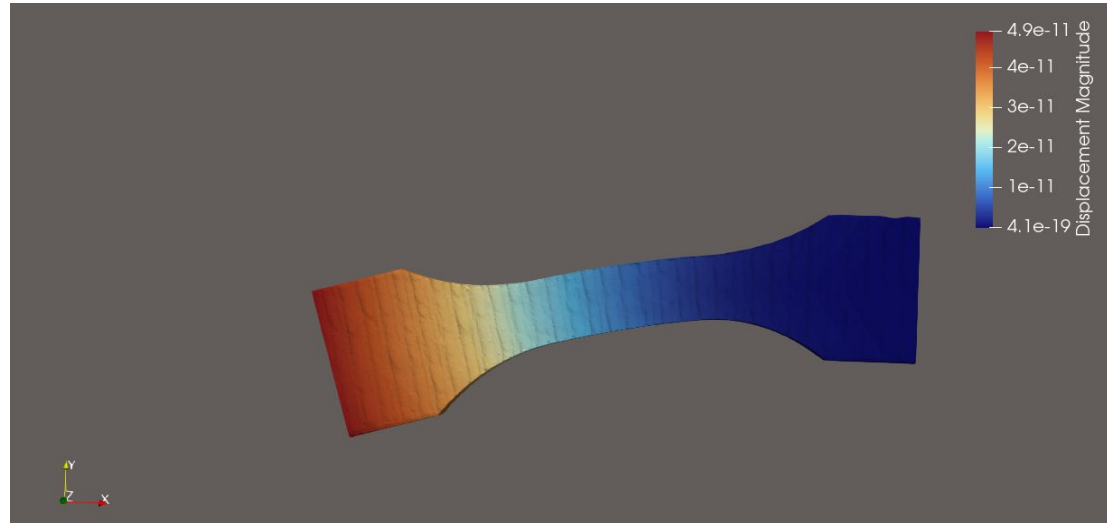


Figure 3.4: Displacement magnitude: left end displaced, right end fixed

T-Shape Specimen - Boundary Condition Setup

Three STL surfaces

STL surface	Physical role	Type	Value
case2_bottom_fixed.stl	Fixed support at base	Dirichlet	[0, 0, 0]
case2_sideA_push.stl	Push down on left arm	Dirichlet (load)	-1.0 (ramped)
case2_sideB_push.stl	Push down on right arm	Dirichlet (load)	-1.0 (ramped)

Figure 3.5: Right Hand Side Push Down (in blue); Bottom Side Fixed (in red)

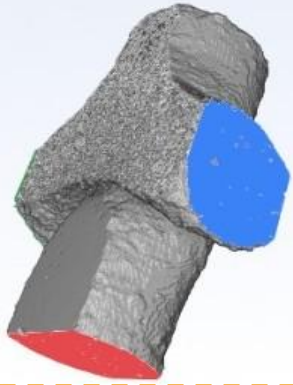
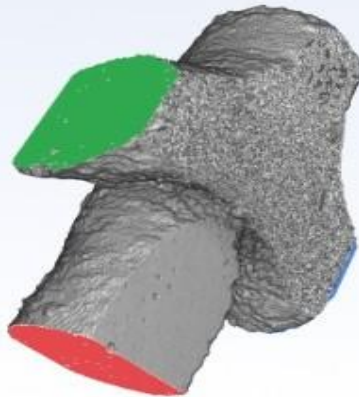


Figure 3.6: Left Hand Side Push Down (in green); Bottom Side Fixed (in red)



Both arm ends pushed downward simultaneously; bottom stem held fixed.

Expected fracture: at the T-junction between stem and arms.

T-Shape Crack Result

- Crack forms in lower stem - at BC surfaces
- Material-weighted penalty working correctly

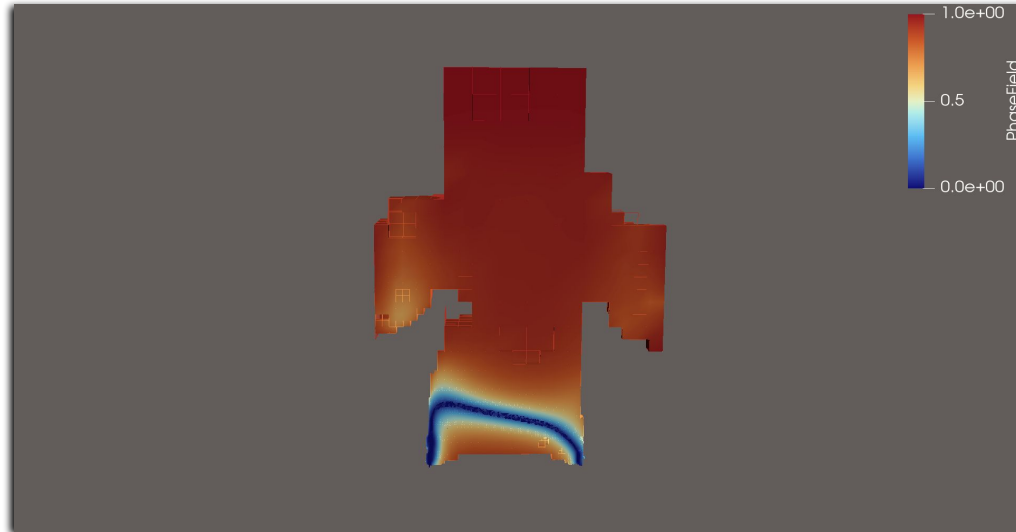


Figure 3.7: Crack band in lower stem · $G_c = 0.01 \text{ kN/mm}$ · Step 100

Phase-Field Fracture: the AT2 model

- Fracture without tracking a crack - just a field, $s \in [0,1]$
- AT2: damage begins the instant stress does - no elastic "free pass"
- Two dials control everything: G_c (energy to break) and l_0 (how sharp the crack is)

Model	σ_c
AT1	$\sqrt{\left(\frac{3 \cdot G_c E}{8 l_0}\right)}$
AT2 ✓	$\sqrt{\left(\frac{27 \cdot G_c E}{256 l_0}\right)}$

Table 4.1 properties of AT1 and AT2 phase-field models

This pipeline uses AT2 throughout - is the standard choice for brittle fracture.

Learning the Parameters: Graphite Four-Point Bending

- Known geometry, known load, known answer
- Built working intuition for I_0 , mesh resolution, and G_c before the real run

0.647 kN

SIMULATION PEAK

0.668 kN

EXPERIMENTAL
TARGET

-3.17%

DEVIATION

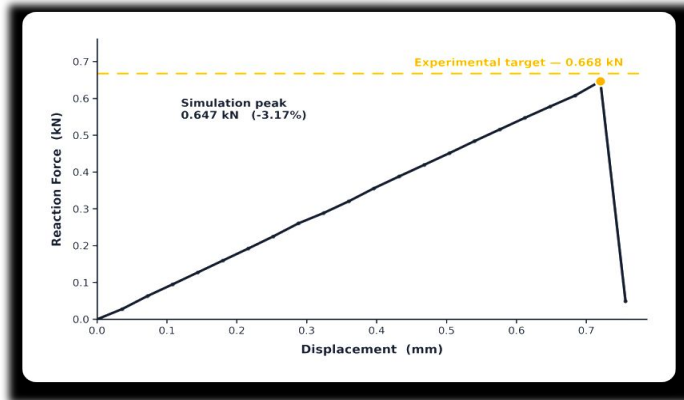


Figure 4.1 Load-displacement graph
($I_0=0.125\text{mm}$, $G_c = 0.118\text{ kN/mm}$, $p=2$)

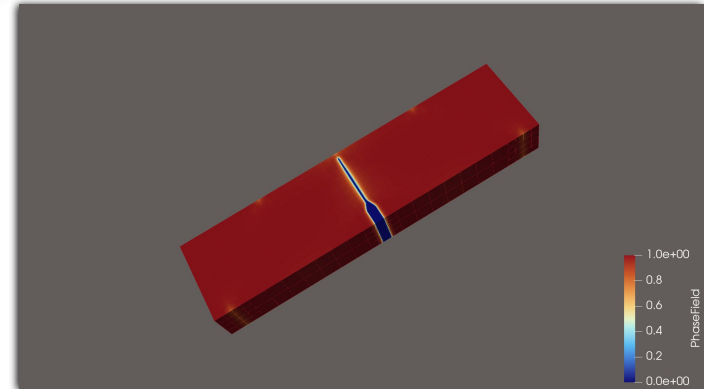


Figure 4.2 Crack nucleates at mid-span and propagates
vertically

Meet the specimen

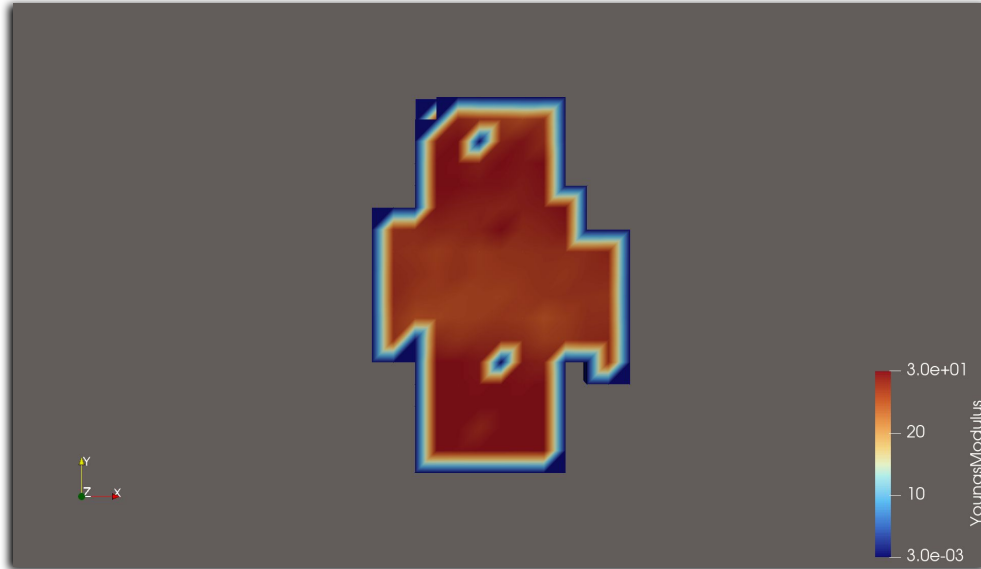


Figure 4.3 Young's Modulus - load step 0 (undeformed)

- Additively manufactured concrete, T-junction geometry - CT scanned, not idealized
- Two real interior voids visible — genuine AM porosity, not a modeling artifact

T-Shape Results

- Real scanned geometry - not an idealized shape
- Displacement: fixed base stays put, loaded arms move
- Damage localizes: arm tips + base of the stem

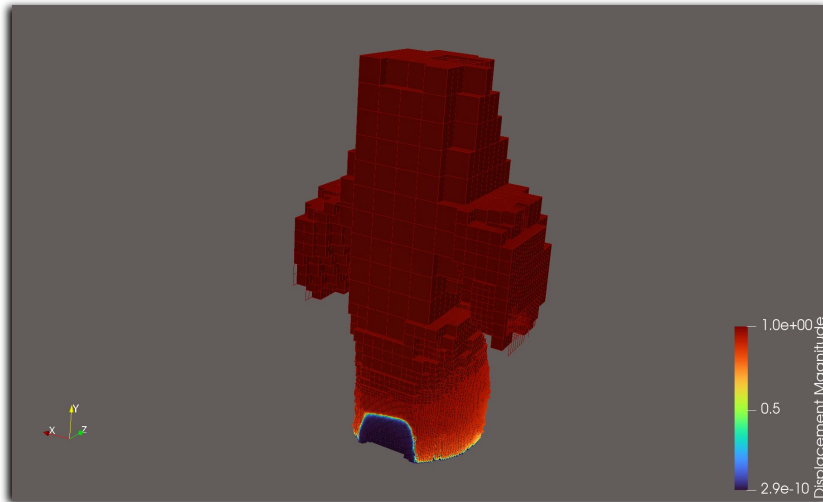


Fig 4.4 Displacement magnitude

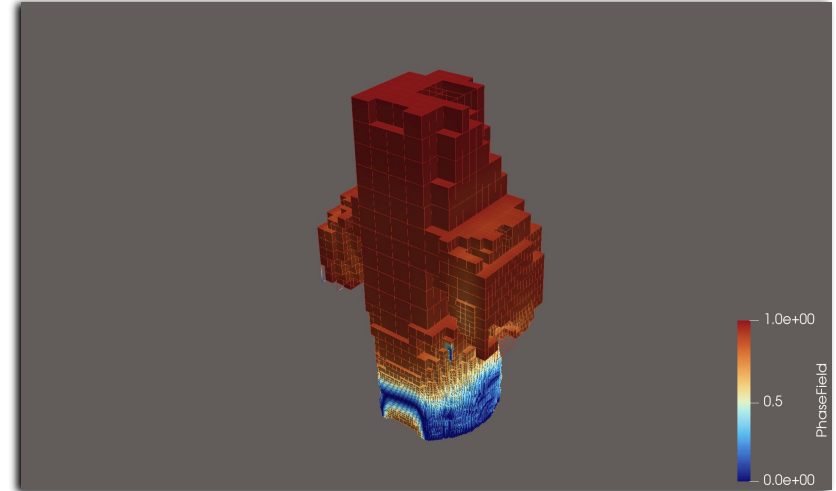


Fig 4.5 Phase field (damage)

Gc Sensitivity Study: Literature-calibrated value

$G_c = 0.0001 \text{ kN/mm}$ (Navidtehrani et al. (2021), cast-concrete AT2) [11] · $n=100$ steps, 1 mm displacement

- No published G_c value exists for AM concrete - this is a deliberate sensitivity study, not a calibration claim

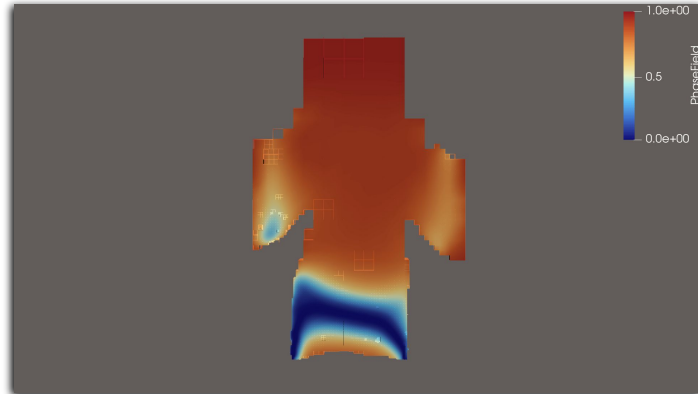


Fig 4.5 Step 20 - damage established

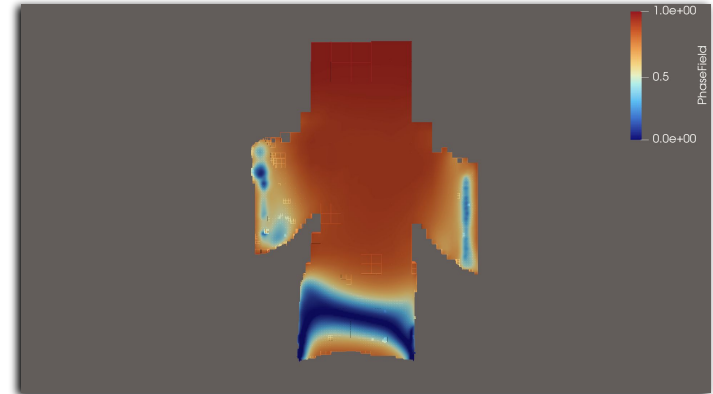


Fig 4.6 Step 100 - pattern holds

Observation: at the literature-matched G_c , damage saturates almost immediately near the base and holds stable through the full run, rather than propagating gradually

Gc Sensitivity Study: Elevated value

$G_c = 0.01 \text{ kN/mm}$ ($\sim 100\times$ literature reference, evolution demonstration) · $n=100$ steps, 1.0mm displacement

- Same T-junction crack origin as the literature-value run - now testing sensitivity to G_c

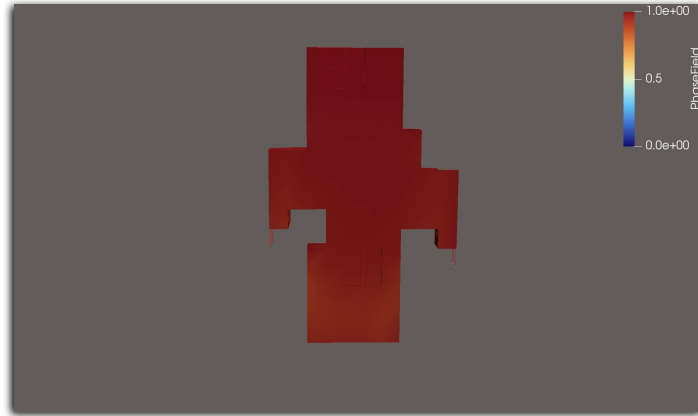


Fig 4.7 Step 20 - damage still diffuse

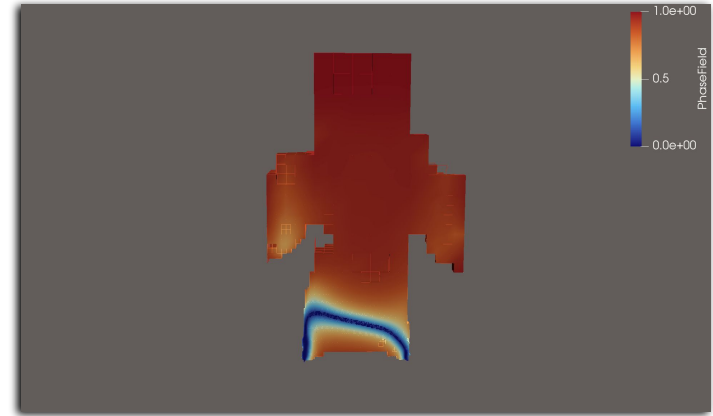


Fig 4.8 Step 100 - crack has propagated

Observation: At an elevated G_c , the crack starts in the same location but visibly propagates by step 100, rather than saturating instantly - a different pattern from the literature-value run in Slide 28.

Interpretation

- Crack initiation location: the same in both runs, regardless of G_c
- G_c controls what happens after initiation, not where it starts:
 - Literature value $\rightarrow$ damage saturates almost immediately, stays stable
 - Elevated value $\rightarrow$ needs more accumulated energy, visibly propagates (step 20–100)
- Two separate behaviors: initiation (likely geometry / boundary-condition driven) vs. propagation (directly controlled by G_c)
- Does not resolve geometry vs. microstructure — that question stays open
- Neither G_c value has an experimental AM-concrete target — a sensitivity study, not a calibration

Conclusion

Achieved

- End-to-end pipeline: CT scan → preprocessed NIfTI → STL boundary conditions → FCM → phase-field simulation → results
- Boundary-condition surfaces extracted directly from CT data and validated through the GUI.
- Boundary conditions correctly enforced on real, CT-derived T-shape geometry
- Generic JSON-driven BC system - any test type configurable without code changes
- Sensitivity behavior characterized across two orders of G_c

Open for Future Work

- AM-concrete-specific G_c calibration (currently anchored to cast-concrete literature)
- Find a way to map the grayscale intensities to proper young's modulus, not just 1 and 0 but with some physical GPa for more natural simulation results.
- Improve GUI automation and usability, including interactive surface selection and automatic validation of exported STL surfaces.

References

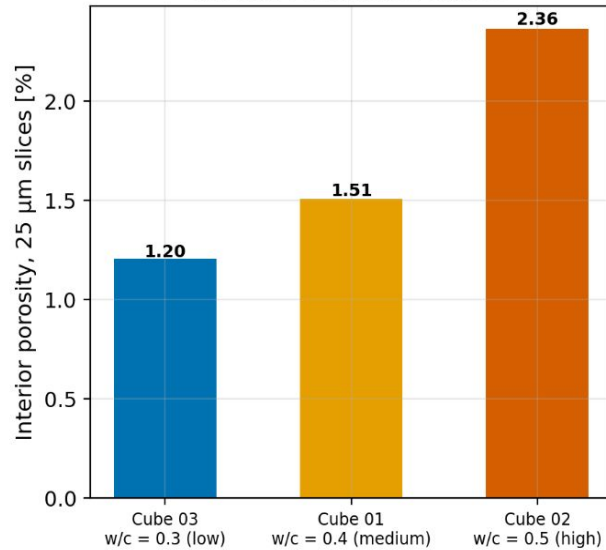
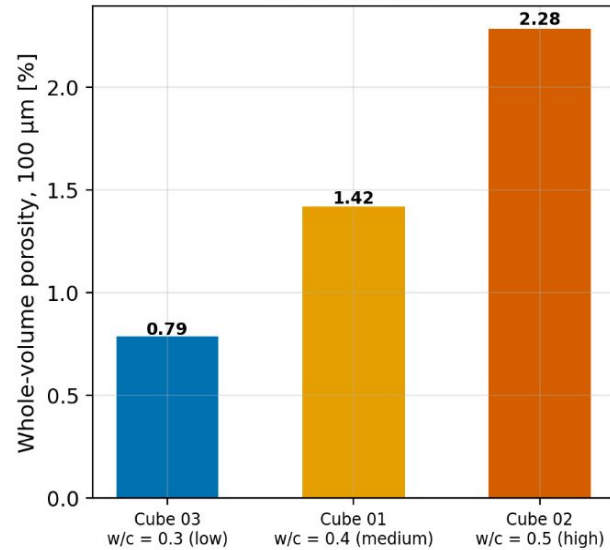
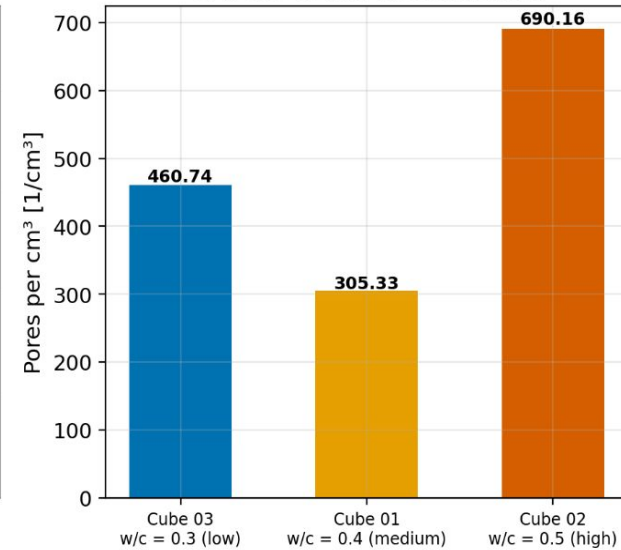
- [1] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66.
- [2] Brooks, R. A., & Di Chiro, G. (1976). Beam hardening in X-ray reconstructive tomography. *Physics in Medicine and Biology*, 21(3), 390–398.
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- [4] Van de Casteele, E., Van Dyck, D., Sijbers, J., & Raman, E. (2004). A model-based correction method for beam hardening artefacts in X-ray microtomography. *Journal of X-ray Science and Technology*, 12(1), 43–57.
- [5] Kittler, J., & Illingworth, J. (1986). Minimum error thresholding. *Pattern Recognition*, 19(1), 41–47.
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References

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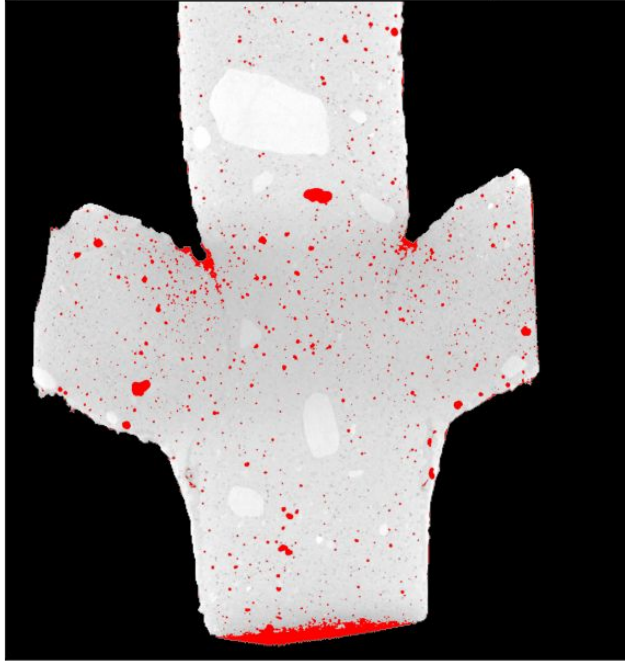
Thank You!

Annex

A: full-resolution slices**B: 3D volume****C: pore count (descriptive)***Figure: Ranking cubes based on porosity analysis*

Annex

pores (red) T=11157 | porosity=3.31%



pores (red) T=8873 | porosity=3.08%

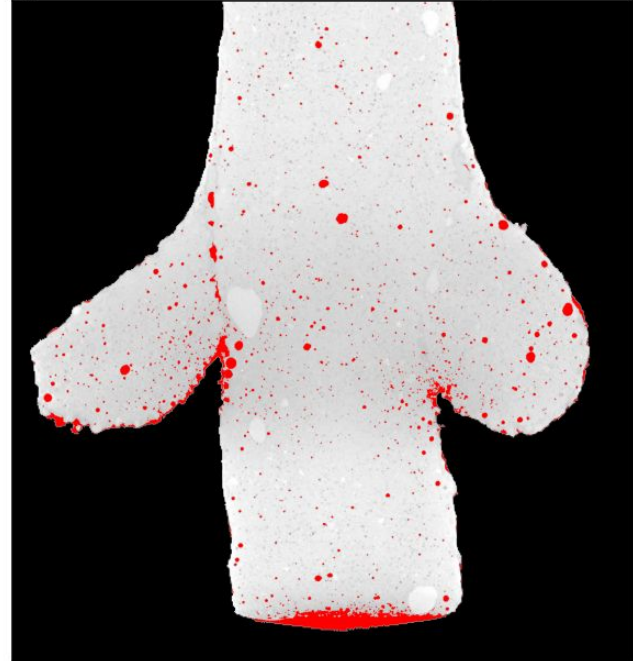


Figure: Pore Segmentation in T-Shape CT Scans

Annex

Full-resolution (50 μm) whole-volume Young's modulus — background removed

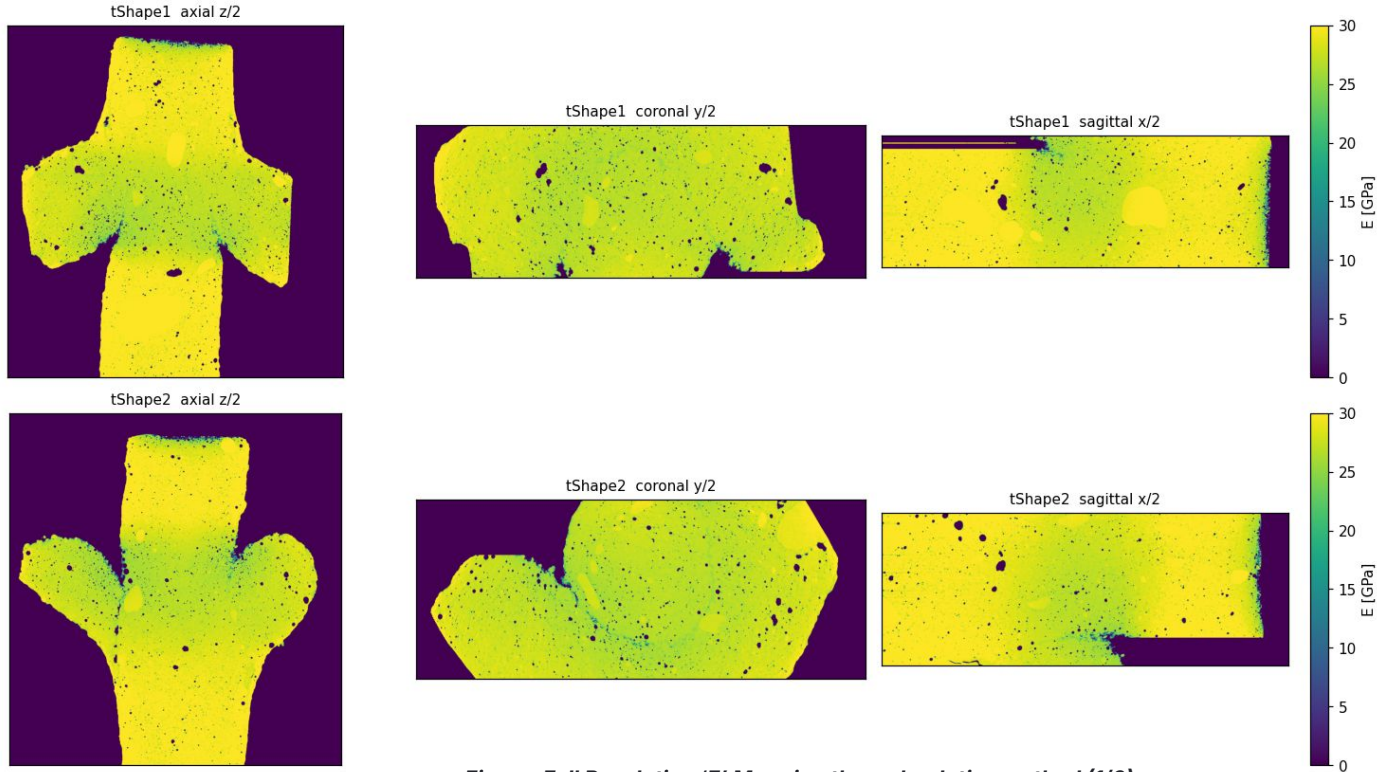


Figure: Full Resolution 'E' Mapping through relative method (1/0)